



Protocol Intelligence Report

Classification

Internal review draft

File: trial_protocol_report.pdf

Structured benchmarking, decision-grade analysis, PubMed evidence enrichment, and audit-ready reporting for clinical trial design review.

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Prepared for clinical development, trial design, and operational planning review.

1. Executive Summary

This report provides a structured benchmarking assessment of the reviewed Phase 3 Interventional protocol targeting Peripheral AVF stenosis. The analysis draws on a matched cohort of **19 precedent studies** retrieved from the ClinicalTrials.gov registry, of which **10 reached completion** and **2 were disrupted** (terminated, suspended, or withdrawn). Evidence strength for this cohort is rated **Limited**.

Across the seven design domains assessed, the reviewed draft shows **notable alignment with disrupted precedent**. The completed-precedent fit score is **38.6%** compared to a disrupted-precedent fit of **64.3%**, yielding a net precedent gap of **-25.7%**. The overall posture classification is: **Closer to disrupted precedent**.

The analysis has surfaced **4 high-priority action(s)** requiring attention before design lock: Allocation — Revisit the allocation strategy or document why the current choice remains defensible.; Masking — Reassess the masking plan and explicitly document any trade-off between feasibility and evidence credibility.

All outputs are intended to support expert review and should be interpreted alongside clinical, statistical, operational, and regulatory judgment. No finding in this report constitutes a regulatory opinion or approval recommendation.

Field	Value
Matched cohort	19 studies
Completed comparators	10
Disrupted comparators	2
Evidence strength	Limited
Completed design fit	38.6%
Disrupted design fit	64.3%
Net precedent gap	-25.7%
Overall posture	Closer to disrupted precedent

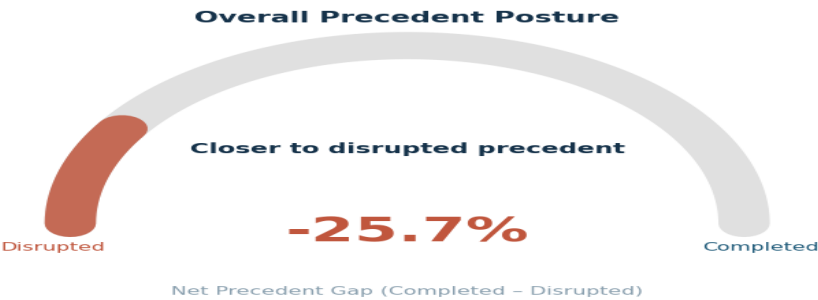


Figure 1. Overall Precedent Posture Gauge — net gap between completed and disrupted design fit.

2. Senior Decision Signals

The following table summarises the five critical decision areas assessed in this benchmark. Each signal should be reviewed by the senior design team before moving to protocol finalisation or governance submission.

Decision Area	Signal	Implication	Evidence
Evidence base	Limited	Confidence is highest when completed and disrupted comparators are both adequately represented.	19 total studies 10 completed 2 disrupted.
Overall posture	Closer to disrupted precedent	This is the clearest top-line read on whether the draft is behaving more like successful or disrupted precedent.	Completed fit 38.6% Disrupted fit 64.3%.
Enrollment plan	Within completed-trial range	Enrollment targets outside the completed range need stronger feasibility justification.	Target 94 Completed IQR 59.2 to 157.0.
Execution exposure	Moderate disruption exposure	Higher disruption prevalence should trigger more explicit mitigation planning.	Disrupted-status share 10.5%.
Protocol completeness	Complete	Missing design fields make the benchmark less trustworthy and reduce report defensibility.	No missing core fields detected.
Lead action	High	Revisit the allocation strategy or document why the current choice remains defensible.	Completed match 50.0% versus disrupted match 100.0%.

3. Executive Action Register

Actions are prioritised as **High** (requires resolution before sign-off), **Medium** (should be addressed before governance), **Monitor** (watch as evidence evolves), or **Preserve** (design choice is well-supported — maintain it). This

run generated **6 action(s)**.

Priority	Category	Action	Recommendation	Evidence
High	Allocation	Challenge	Revisit the allocation strategy or document why the current choice remains defensible.	Completed match 50.0% versus disrupted match 100.0%.
High	Masking	Challenge	Reassess the masking plan and explicitly document any trade-off between feasibility and evidence credibility.	Completed match 20.0% versus disrupted match 50.0%.
High	Overall design posture	Challenge	Rework the draft so it aligns more closely with completed-trial precedent before sign-off.	Completed fit 38.6% versus disrupted fit 64.3%.
High	Primary Purpose	Challenge	Tighten the primary purpose framing so the design and endpoint package read as coherent.	Completed match 50.0% versus disrupted match 100.0%.
Medium	Study Type	Challenge	Pressure-test whether the study type is the strongest route for the program decision.	Completed match 60.0% versus disrupted match 100.0%.
Monitor	Evidence strength	Monitor	Treat the benchmark output as directional until the comparator evidence base is broadened.	Total matched studies 19; completed 10; disrupted 2.

4. Reviewed Protocol Profile

The following fields were extracted from the uploaded protocol document and reviewed by the study team. Extraction confidence: **medium**. Profile status: **Reviewed**.

Field	Value
Title	Randomised clinical trial comparing drug-coated balloon to plain balloon for all peripheral AVF stenosis
Condition	Peripheral AVF stenosis
Sponsor	Boston Scientific
Study Type	Interventional
Phase	Phase 3
Planned Enrollment	94
Arms Count	2
Allocation	Randomized
Masking	Single
Intervention Model	Parallel Assignment
Primary Purpose	Treatment
Comparator	Plain balloon angioplasty
Target Population	Renal failure patients with malfunctioning AVFs
Geography Focus	Singapore
Start Date	Not provided
Completion Date	Not provided

Endpoint Focus	Not provided
Primary Endpoints	Access circuit primary patency at 6 months
Secondary Endpoints	Access circuit primary patency at 12 months, Access circuit assisted primary patency at 6 and 12 months, Access circuit secondary patency at 6 and 12 months, Number of repeat interventions at 12 months, Procedural complication rates, Mortality rates at 12 months and up to 5 years

5. Precedent Posture Analysis

Overall precedent posture: **Closer to disrupted precedent** — based on 19 matched studies. The draft is closer to disrupted-trial precedent (fit 64.3%) than to completed-trial precedent (fit 38.6%), a deficit of 25.7%. This is a meaningful risk signal. The team should review the domain-level breakdown to identify which design choices are driving the pattern.

6. Design Domain Alignment

Study Type shows a risk signal: completed match 60.0% vs disrupted match 100.0%. Study type changes how decision-makers interpret trial evidence and fit-for-purpose value. The team should review this domain explicitly before design lock.

Allocation shows a risk signal: completed match 50.0% vs disrupted match 100.0%. Allocation strategy affects comparability, bias control, and governance confidence. The team should review this domain explicitly before design lock.

Masking shows a risk signal: completed match 20.0% vs disrupted match 50.0%. Masking shapes bias control, endpoint credibility, and operational burden. The team should review this domain explicitly before design lock.

Primary Purpose shows a risk signal: completed match 50.0% vs disrupted match 100.0%. Primary purpose anchors endpoint expectations and how the study intent is judged. The team should review this domain explicitly before design lock.

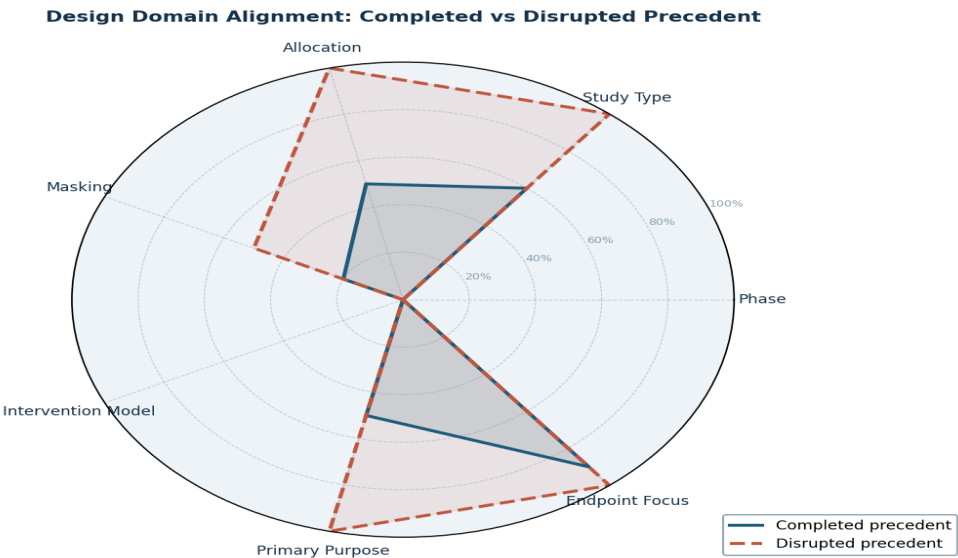


Figure 2. Design Domain Alignment Radar — completed precedent (teal filled) vs disrupted precedent (red dashed). Larger filled area = stronger alignment with successful trials.

Design Alignment Risk Signal Summary				
Domain	Completed %	Disrupted %	Gap (C-D)	Signal
Phase	0.0%	0.0%	+0.0%	Mixed
Study Type	60.0%	100.0%	-40.0%	Closer to Disrupted
Allocation	50.0%	100.0%	-50.0%	Closer to Disrupted
Masking	20.0%	50.0%	-30.0%	Closer to Disrupted
Intervention Model	0.0%	0.0%	+0.0%	Mixed
Primary Purpose	50.0%	100.0%	-50.0%	Closer to Disrupted
Endpoint Focus	90.0%	100.0%	-10.0%	Mixed

Figure 3. Domain Alignment Risk Signal Summary — green = closer to completed precedent; red = closer to disrupted.

7. Design Differential Matrix

This matrix shows, domain by domain, the match rate the protocol has with completed and disrupted precedent, the net gap between them, and the signal classification. The 'Why It Matters' context helps reviewers prioritise which gaps need explicit narrative justification.

Domain	Protocol Choice	Completed %	Disrupted %	Gap %	Signal
Phase	Phase 3	0.0	0.0	0.0	Weak precedent signal
Study Type	Interventional	60.0	100.0	-40.0	Closer to disrupted precedent
Allocation	Randomized	50.0	100.0	-50.0	Closer to disrupted precedent
Masking	Single	20.0	50.0	-30.0	Closer to disrupted precedent
Intervention Model	Parallel Assignment	0.0	0.0	0.0	Weak precedent signal
Primary Purpose	Treatment	50.0	100.0	-50.0	Closer to disrupted precedent
Endpoint Focus	Other	90.0	100.0	-10.0	Mixed precedent

8. Enrollment Benchmark

Enrollment benchmarking shows that the protocol target of 94 falls **within the completed-trial IQR (59–157)**, which is a neutral signal. Execution and site selection will be the primary feasibility levers. Completed-trial median enrollment was 95. Disrupted trials had a median of 61 (IQR 34–88), suggesting that enrollment sizing alone is not a reliable differentiator between successful and disrupted studies — operational and design factors matter more.

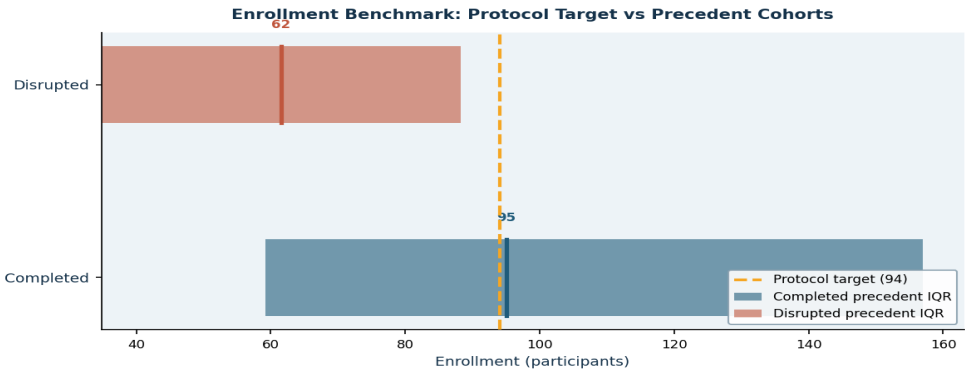


Figure 4. Enrollment Benchmark — IQR boxes show 25th–75th percentile range; centre line is the median; amber dashed = protocol target.

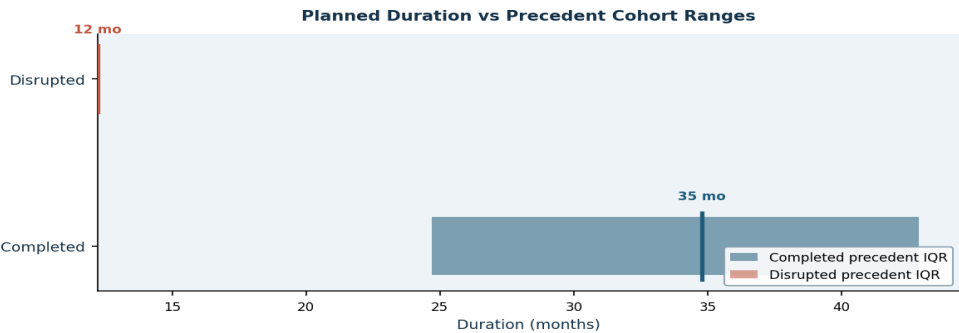


Figure 5. Planned Duration vs Precedent Cohort Ranges — amber dashed line = protocol planned duration.

9. Comparator Cohort Definition

The following table describes how the comparator cohort was constructed. Only studies matching the clinical condition specified in the protocol were included. Status classification (completed vs disrupted) follows ClinicalTrials.gov status fields.

Metric	Value
Clinical condition	Peripheral AVF stenosis
Matched trial count	19
Completed comparators	10
Disrupted comparators	2
Active comparators	2
Evidence strength	Limited
Risk-status share	10.5%
Median site count	1.0
Median country count	1.0

10. Success vs Disruption Benchmark

This table presents, for each design parameter, the protocol's value alongside the comparable range from completed and disrupted trials. The signal column classifies the protocol's position relative to completed-trial precedent.

Domain	Protocol Draft	Completed Precedent	Disrupted Precedent	Signal
Enrollment	94	Median 95.0 IQR 59.2 to 157.0	Median 61.5 IQR 34.8 to 88.2	Within completed-trial range
Planned Duration	Not provided	Median 34.8 mo IQR 24.7 to 42.9 mo	Median 12.2 mo IQR 12.2 to 12.2 mo	Insufficient evidence
Phase	Phase 3	0.0%	0.0%	Weak precedent signal
Study Type	Interventional	60.0%	100.0%	Closer to disrupted precedent
Allocation	Randomized	50.0%	100.0%	Closer to disrupted precedent

Domain	Protocol Draft	Completed Precedent	Disrupted Precedent	Signal
Masking	Single	20.0%	50.0%	Closer to disrupted precedent
Intervention Model	Parallel Assignment	0.0%	0.0%	Weak precedent signal
Primary Purpose	Treatment	50.0%	100.0%	Closer to disrupted precedent
Endpoint Focus	Other	90.0%	100.0%	Mixed precedent

11. Decision Scorecard

Quantitative scorecard of all computed metrics. These numbers underpin the signals and recommendations in this report. Any metric showing 'Not available' reflects a data gap in either the protocol or the comparator cohort.

Metric	Value
Matched cohort size	19
Completed comparator count	10
Disrupted comparator count	2
Evidence strength	Limited
Completed precedent fit (%)	38.6
Disrupted precedent fit (%)	64.3
Net precedent gap (%)	-25.7
Overall posture	Closer to disrupted precedent
Enrollment target	94
Completed enrollment IQR	59.2 to 157.0
Disrupted enrollment IQR	34.8 to 88.2
Completed median duration (months)	34.8
Disrupted median duration (months)	12.2
Median site count	1.0
Median country count	1.0
Risk-status share (%)	10.5

12. Endpoint Category Evidence

Endpoint category distribution reveals what types of evidence completed and disrupted trials have relied upon in this condition. The protocol's current endpoint focus is highlighted. Aligning with completed-trial endpoint patterns is associated with stronger design credibility.

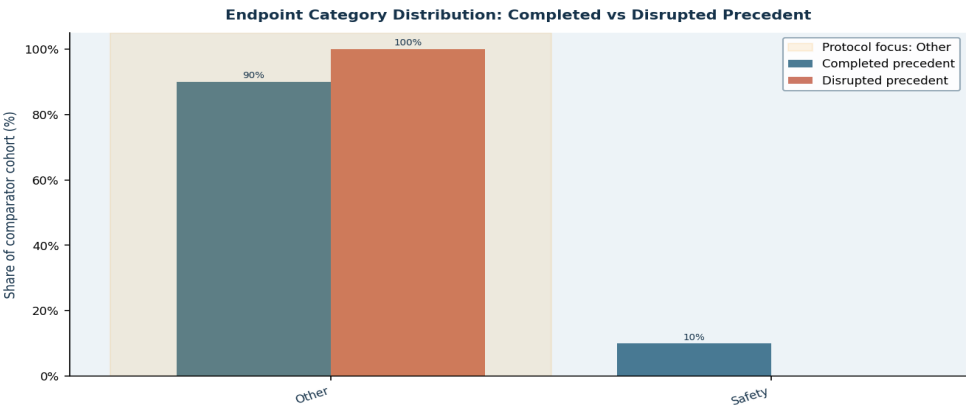


Figure 6. Endpoint Category Distribution — completed precedent (teal) vs disrupted precedent (red). Amber shading = protocol endpoint focus.

Endpoint Category	Completed Share (%)	Disrupted Share (%)	Net Gap (%)
Other	90.0	100.0	-10.0
Safety	10.0	0.0	10.0

13. PubMed Literature Evidence

2 peer-reviewed article(s) were retrieved from PubMed/MEDLINE via the NCBI E-utilities free API to provide literature context for this trial design. All citations below are fully traceable — each entry includes the PubMed ID (PMID), authors, journal, year of publication, and the query used to retrieve it. The abstract excerpts are provided to help the reviewer judge relevance quickly. Full articles can be accessed at [pubmed.ncbi.nlm.nih.gov/\[PMID\]](https://pubmed.ncbi.nlm.nih.gov/[PMID]).

PMID	Authors	Year	Journal	Title	Endpoint Focus
36758984	Lindhard K et al.	2023	Nephrology, dialysis, transplantati	Far infrared treatment on the arteriovenous fistula induces changes in sVCA	General
12566672	Lemson S et al.	2002	Blood purification	Hemodynamics of venous cuff interposition in prosthetic arteriovenous fistu	General

Abstract Excerpts

Far infrared treatment on the arteriovenous fistula induces changes in sVCAM and sICAM in patients on hemodialysis. — Lindhard K, Jensen BL, Pedersen BL | Nephrology, dialysis, transplantation : official publication of the European Dialysis and Transplant Association - European Renal Association (2023) PMID: 36758984 | Query: "Peripheral AVF stenosis" AND (clinical trial[pt] OR randomized controlled trial[pt])

There is a substantial risk of developing stenosis and dysfunction in the arteriovenous fistula (AVF) in patients on hemodialysis (HD). Far infrared radiation (FIR) is a non-invasive local intervention with a potentially beneficial effect on AVF patency. The underlying mechanism is not clear. It was hypothesized that a single FIR treatment reduces factors of inflammation and promotes endothelial vasodilators in the AVF. Forty HD patients with an AVF were included in an open-label intervention study. Patients were randomized to receive either FIR (FIR group) or no FIR (control group). Blood sam...

Hemodynamics of venous cuff interposition in prosthetic arteriovenous fistulas for hemodialysis. — Lemson S, Tordoir JH, Ezzahiri R | Blood purification (2002) PMID: 12566672 | Query: "Peripheral AVF stenosis" AND (clinical trial[pt] OR randomized controlled trial[pt])

The durability of prosthetic arteriovenous fistulas (AVF) for hemodialysis is jeopardized by thrombotic occlusions due to intimal hyperplastic stenoses. In arterial reconstructive surgery, peripheral arterial bypasses with prosthetic material benefit from a venous cuff at the distal anastomosis. Therefore, a study was performed to assess the effect of a venous cuff at the venous anastomosis of PTFE graft AVFs in terms of stenosis development, hemodynamics and patency rates. A subset of 40 patients from a multicenter study were enrolled into the study, of which 20 patients were randomized for v...

14. Comparative Narrative

Matched 19 studies for Peripheral AVF stenosis, including 10 completed comparators and 2 disrupted comparators. Top-line posture: Closer to disrupted precedent (completed fit 38.6%; disrupted fit 64.3%). Enrollment position: Within completed-trial range. Protocol target 94 | completed-trial IQR 59.2 to 157.0 | disrupted-trial IQR 34.8 to 88.2. Most important design differential: Allocation shows 50.0% completed-trial match versus 100.0% disrupted-trial match. Lead action: Revisit the allocation strategy or document why the current choice remains defensible. Interpretation note: this output is intended to support senior design review, not replace clinical, statistical, operational, or regulatory judgment.

15. Recommendation Detail

Full detail on all recommendations generated by this analysis run. Each recommendation includes a rationale derived from the precedent data and a specific evidence statement referencing the underlying statistics.

Priority	Category	Recommendation	Rationale	Evidence
High	Allocation	Revisit the allocation strategy or document why the current choice remains defensible.	Allocation strategy affects comparability, bias control, and governance confidence.	Completed match 50.0% versus disrupted match 100.0%.
High	Masking	Reassess the masking plan and explicitly document any trade-off between feasibility and evidence credibility.	Masking shapes bias control, endpoint credibility, and operational burden.	Completed match 20.0% versus disrupted match 50.0%.
High	Overall design posture	Rework the draft so it aligns more closely with completed-trial precedent before sign-off.	Across the available design fields, the current draft is behaving more like disrupted precedent than completed precedent.	Completed fit 38.6% versus disrupted fit 64.3%.
High	Primary Purpose	Tighten the primary purpose framing so the design and endpoint package read as coherent.	Primary purpose anchors endpoint expectations and how the study intent is judged.	Completed match 50.0% versus disrupted match 100.0%.
Medium	Study Type	Pressure-test whether the study type is the strongest route for the program decision.	Study type changes how decision-makers interpret trial evidence and fit-for-purpose value.	Completed match 60.0% versus disrupted match 100.0%.
Monitor	Evidence strength	Treat the benchmark output as directional until the comparator evidence base is broadened.	The matched cohort is still thin for a strong success-versus-disruption readout.	Total matched studies 19; completed 10; disrupted 2.

16. Comparator Exemplars

The table below lists representative studies from the matched cohort, separated by comparator lens (completed precedent, disrupted precedent, active context). These studies can be reviewed directly on ClinicalTrials.gov using the NCT ID provided.

Comparator Lens	Status	NCT ID	Title	Enrollment	Sponsor
Completed Precedent	Completed	NCT05372952	An Investigator Initiated Trial to Evaluate the Safety and Feasibility of the DyNETIC-35 Stent for Iliac Lesions Via a Trans-radial Approach	25	Klinikum Arnsberg
Completed Precedent	Completed	NCT03270358	OSTéopontin as a Marker Of StenoSIS - OSMOSIS	76	Centre Hospitalier Universitaire de Nice
Completed Precedent	Completed	NCT02335099	Determine the Safety/Efficacy of Ticagrelor for Maintaining Patency of Arterio-Venous Fistulae Created for Hemodialysis	54	University of Virginia
Completed Precedent	Completed	NCT04226599	A Safety and Efficacy Study of Hemodialysis Arteriovenous Fistulae Stenosis Treated With DissolveAV	220	DK Medical Technology (Suzhou) Co., Ltd.
Disrupted Precedent	Terminated	NCT02224235	COBRA SHIELD OCT Study	8	CeloNova BioSciences, Inc.
Disrupted Precedent	Terminated	NCT02913274	Arteriovenous Fistulae: Drug-eluting Balloon Angioplasty	115	Rennes University Hospital
Active Context	Recruiting	NCT06553443	Randomised Clinical Trial Comparing Drug-coated Balloon to Plain Balloon for All Peripheral AVF Stenosis	94	Singapore General Hospital
Active Context	Active Not Recruiting	NCT04540302	The Merit WRAPSODY AV Access Efficacy Study (WAVE)	357	Merit Medical Systems, Inc.

18. Audit Trail and Provenance

Every action taken in this session is recorded below with a UTC timestamp, actor, artifact type, and full detail string. This trail supports governance review, regulatory readiness, and reproducibility audits.

2026-03-19 08:02 UTC Extract Protocol Profile
Actor: System Artifact: Document ID: Prot_000.pdf
Extracted protocol profile from Prot_000.pdf.
2026-03-19 08:02 UTC Review Protocol Profile
Actor: User Artifact: Protocol Profile
Reviewer updated and confirmed protocol profile fields.
2026-03-19 08:02 UTC Build Comparable Cohort
Actor: System Artifact: Comparison Cohort
Built registry comparison cohort for Peripheral AVF stenosis.
2026-03-19 08:03 UTC Fetch Pubmed Evidence
Actor: System Artifact: Literature Evidence ID: Peripheral AVF stenosis Metadata: article_count: 2, endpoint_focus: any
Retrieved 2 PubMed articles for 'Peripheral AVF stenosis'.
2026-03-19 08:04 UTC Export Pptx Slides
Actor: System Artifact: Slides ID: trial_protocol_slides.pptx Metadata: pubmed_articles: 2
Generated trial_protocol_slides.pptx.
2026-03-19 08:07 UTC Export Pdf Report
Actor: System Artifact: Report ID: trial_protocol_report.pdf Metadata: pubmed_articles: 2
Generated trial_protocol_report.pdf.

19. Methodology Notes

Source trials were retrieved from the ClinicalTrials.gov public registry (API v2) and normalised into a comparator cohort filtered to the protocol's clinical condition. Trials were classified as **completed** (status: COMPLETED) or **disrupted** (status: TERMINATED, SUSPENDED, or WITHDRAWN) and used as two distinct analytical lenses. Design alignment was assessed across seven domains: Phase, Study Type, Allocation, Masking, Intervention Model, Primary Purpose, and Endpoint Focus. Alignment scores represent the proportion of comparator trials sharing the protocol's design choice in each domain.

Protocol fields were extracted using deterministic heuristic parsing and, where configured, an LLM-assisted structured extraction step. PubMed evidence was retrieved via the NCBI E-utilities API (free access, no key required) using a condition-and-design-context query. All LLM calls are logged with model name and confidence metadata.

All outputs in this report are intended to support expert review and informed decision-making. They do not constitute a

regulatory opinion, clinical recommendation, or approval decision. The report should be reviewed alongside clinical, statistical, operational, and regulatory expertise appropriate to the development programme.